

Figure 1.5 Blockchain node and application stack

> **DEFINITION** An *account* represents a unique identity for a transacting entity. An account is needed to initiate a transaction.

A blockchain node hosts the elements represented by the stack in figure 1.5. It serves as a foundation for your blockchain application development.

Let's start from the bottom and move up. The lower two levels are the standard hardware and software of most computing systems. The next level up is the blockchain protocol level: it houses the components of the blockchain, but you won't program at this level. The next layer hosts the application logic. This layer is where you solve problems like access control to data and code functions for validation, verification, and recording. The top layer is the user-facing interface where web (or enterprise) programming is done, such as with HTML, JavaScript, and associated frameworks. These elements form the Dapp and its user interface (UI) layer.

A blockchain application is not a single-user application, unlike a handheld game or an income tax calculator. It typically connects a large number of participants through its network of nodes. Each node can host multiple accounts to identify the different customers it services. A node can also